Article

Data-Driven RGB-LED Sensor Fusion for Spectral Reflectance Reconstruction: An Interpretable Framework with D-Optimal Design and Bayesian Uncertainty Quantification

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Abstract

Background: Spectral reflectance reconstruction from RGB measurements is an ill-posed inverse problem: three broadband channels are insufficient for stable inversion in most metrological applications. Adding a small number of carefully chosen LED channels mitigates this ill-posedness at negligible hardware cost. **Methods:** We propose a compact sensor-fusion pipeline that couples (i) a PCA latent basis with data-driven dimension selection, (ii) D-optimal LED wavelength selection that maximizes the log-determinant of the Fisher information matrix in latent-coefficient space, and (iii) Bayesian reconstruction with closed-form posterior uncertainty. Sensing quality is quantified through the condition number $\kappa(A\Phi)$ and $\log\det((A\Phi)^\top A\Phi)$, both independent of the test sample and of the choice of estimator. **Results:** On a controlled thin-film benchmark, D-optimal fusion with $m_{\mathrm{LED}} = 4$ reduces $\kappa(A\Phi)$ from 2.687 to 1.251, raises log det from -10 to -6, and achieves mean $\Delta E_{00} = 1.72$ versus 2.31 for RGB-only (under shot noise; $p < 0.01$, paired Wilcoxon). Results are replicated on Gaussian broad-band and artist-paint proxy libraries. A LED-budget sensitivity analysis shows diminishing returns beyond $m_{\mathrm{LED}} = 4$, providing a principled stopping criterion. **Conclusions:** The framework occupies a well-defined niche as an interpretable, reproducible baseline that provides sensing-geometry transparency and calibrated uncertainty envelopes not available from neural estimators. Code and data are provided for benchmarking.

Keywords: spectral reflectance reconstruction; RGB-LED fusion; color metrology; D-optimal design; Bayesian estimation; Fisher information; uncertainty quantification; PCA; low-cost instrumentation

1. Introduction

Accurate measurement of spectral reflectance is foundational in digital imaging, industrial color inspection, coatings, cultural heritage imaging, thin-film monitoring, and appearance engineering [1]. The same need for low-cost, portable spectral sensing arises in precision agriculture and food quality inspection [2], biomedical and dermatological reflectance imaging [3], waste sorting and recycling [4], textile and dye quality control [5], and graphic-arts and fine-art color reproduction [6]. The reflectance function mediates both physical interpretation and perceived color, but the devices that measure it reliably—full-range spectrophotometers and multispectral imagers—can be expensive, slow, or operationally restrictive.

Received:
Revised:
Accepted:
Published:

 https://doi.org/10.3390/technologies1010000

A practical alternative is to infer reflectance from low-dimensional measurements. RGB cameras are inexpensive and widely available, yet their three broadband channels compress the underlying spectral signal too severely for stable inversion in many applications. A natural compromise is to add a small number of narrow-band illuminants: LED modules are low-cost, compact, and synchronizable with standard cameras [7,8].

The mathematical challenge is that reflectance reconstruction remains an ill-posed inverse problem after discretization: the unknown is high-dimensional while the measurement vector is short. Prior work has shown that Bayesian and Wiener-type estimators regularize this inversion effectively [9,10] and that neural estimators can further improve accuracy on large datasets [11,12]. What has received less attention is the question of which LED channels to add.

This paper answers that question through D-optimal design and shows that channel selection affects both RMSE and the geometric structure of the inverse problem in ways that are measurable independently of the estimator. The central object is the effective sensing operator $A\Phi$, where A collects RGB and LED channel responses and Φ is a learned spectral basis. This viewpoint makes three issues explicit and separable: (i) the quality of the latent representation, controlled by Φ and the latent dimension K; (ii) the information content of the chosen LED wavelengths, controlled by the D-optimal criterion; and (iii) the numerical stability of the inverse problem, quantified by $\kappa(A\Phi)$ independently of the estimator.

1.1. Novelty and Scope

Three aspects of the present framework are new relative to prior work. First, we integrate D-optimal LED selection directly into the spectral latent-basis pipeline, treating wavelength choice as part of the inference problem rather than a hardware decision. Prior work in RGB-LED fusion [7,8] selects channels empirically; the present work selects them to maximize the Fisher information determinant given a specific latent representation.

Second, we provide a LED-budget sensitivity analysis (Section 4.5) that gives practitioners a principled stopping criterion for m_{LED}.

Third, we combine Fisher-information and conditioning diagnostics with spectral RMSE and colorimetric ΔE_{00} metrics, all reported with bootstrap confidence intervals and paired Wilcoxon tests, into a unified evaluation scaffold that can be applied to any future estimator without modification.

This paper makes five claims, explicitly no more. (C1) Well-chosen LED channels improve an RGB-only sensing architecture, with gains visible in RMSE, ΔE_{00}, and the geometry of the inverse problem. (C2) D-optimal design provides a tractable, principled criterion for channel selection given a target spectral family. (C3) Fisher-information and conditioning diagnostics quantify sensing quality independently of the test sample and of the estimator. (C4) Bayesian inference provides calibrated uncertainty envelopes in all noise regimes and, under heteroscedastic shot noise, a statistically significant point-estimation improvement over ridge regression ($p < 0.01$, paired Wilcoxon). (C5) The framework constitutes a reproducible evaluation scaffold with fixed splits, matched budgets, and geometry diagnostics, enabling fair comparison of any future estimator.

2. Related Work

2.1. Low-Dimensional Spectral Models

The low dimensionality of natural reflectance spectra is classical [1]. Linear subspace models, PCA representations, and Wiener-type estimators exploit this structure [9,10]. The present paper extends this line by treating the latent dimension K as a model-selection variable (Section 3.2) rather than a fixed hyperparameter, and by coupling the latent model directly to the D-optimal sensor design.

2.2. Multispectral and Mixed-Channel Architectures

Mixed architectures combining RGB cameras with additional illuminants recover part of the information content of multispectral systems at a fraction of the hardware cost [7,8]. Previous work has not, however, formalized LED selection as a Fisher-information optimization problem in latent space. The present paper fills this gap.

2.3. Bayesian and Uncertainty-Aware Reconstruction

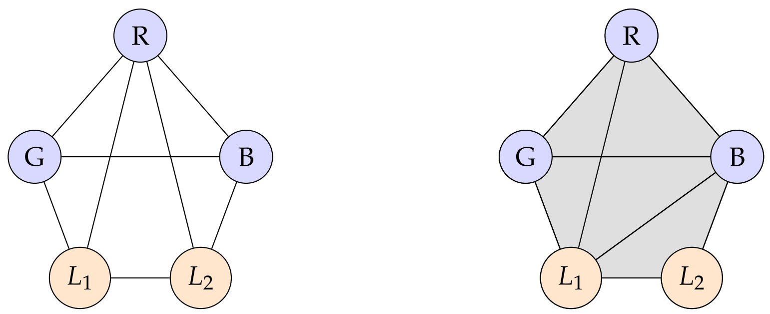
The Bayesian posterior mean coincides with ridge regression under isotropic Gaussian priors and homoscedastic noise [13]; the equivalence breaks under heteroscedastic shot noise, where the likelihood adapts per-channel regularization to signal intensity. This distinction is theoretically established in Section 3.6 and empirically confirmed in Section 4.9.

2.4. Learning-Based Spectral Reconstruction

Recent methods treat spectral reconstruction as an end-to-end learning problem [11,12]. These methods improve accuracy on large datasets but treat the sensor as fixed hardware, providing no geometric insight into channel informativeness. The present framework is complementary: it provides the sensing-design scaffold on which learning-based methods can be evaluated fairly and combined with principled hardware choices.

2.5. Topological and Higher-Order Representations of Channel Interactions

The D-optimal criterion of Definition 1 is a pairwise criterion: $\det((A_{\mathcal{S}}\Phi)^\top(A_{\mathcal{S}}\Phi))$ is built from the Gram matrix of channel responses, so it aggregates only two-channel inner products and cannot directly represent redundancy or synergy that appears jointly among three or more channels at once. Topological deep learning (TDL) generalizes graph-based message passing to simplicial and cell complexes, in which higher-order simplices explicitly carry features and relations defined on k-tuples of nodes rather than only on pairs [14]; message-passing simplicial networks show that this strictly increases expressive power relative to graph-based (pairwise) message passing [15]. This suggests a natural higher-order generalization of Definition 1: representing the candidate library $\mathcal{L}$ together with the RGB channels as the vertex set of a simplicial complex whose k-simplices are weighted by the informativeness of the corresponding $(k+1)$-channel subset, of which the pairwise Fisher-information Gram matrix in Equation (5) is only the 1-skeleton. Figure 1 illustrates the two views schematically.



(a) Pairwise (graph) view
edges $\leftrightarrow$ entries of $(A\Phi)^\top(A\Phi)$

(b) Higher-order (simplicial complex) view
filled triangles $\leftrightarrow$ 3-channel joint terms

Figure 1. Schematic contrast between the pairwise D-optimal criterion used in this paper and a higher-order, topological deep learning generalization. (**a**) The Fisher information Gram matrix $(A\Phi)^\top(A\Phi)$ of Equation (5) only weighs edges between channel pairs. (**b**) A simplicial complex over the same channel set additionally weighs 2-simplices (filled triangles), representing three-channel joint informativeness not reducible to pairwise terms [14,15]. This is a conceptual illustration of a future direction (Section 5.4), not an implemented estimator.

Because this higher-order structure is not implemented or benchmarked in the present paper, it is carried forward as a future direction in Section 5.4 rather than as a fifth claim in Section 1.1.

3. Materials and Methods

3.1. Forward Measurement Model

Let $R(\lambda)$ denote spectral reflectance on $[\lambda_{\min}, \lambda_{\max}]$. For channel i with illumination $E_i(\lambda)$ and detector sensitivity $Q_i(\lambda)$:

$$s_i = \int_{\lambda_{\min}}^{\lambda_{\max}} E_i(\lambda)R(\lambda)Q_i(\lambda)\,d\lambda. \tag{1}$$

Discretizing on a 61-point grid (400–700 nm, 5 nm step) yields the matrix model

$$s = AR + \epsilon, \tag{2}$$

where $A \in \mathbb{R}^{m\times n}$ is the sensing matrix and ϵ is measurement noise. The first three rows of A correspond to RGB channels and the remaining m_{LED} rows to LED channels.

3.2. Data-Driven Latent Representation and Dimension Selection

Because $m \ll n$, direct inversion of (2) is ill-posed. Reflectance is projected onto a learned low-dimensional subspace:

$$R \approx \Phi\alpha = \sum_{k=1}^{K} \alpha_k \phi_k. \tag{3}$$

where $\Phi \in \mathbb{R}^{n\times K}$ is the PCA basis and $\alpha \in \mathbb{R}^K$ are latent coefficients. Substituting into (2):

$$s = A\Phi\alpha + \epsilon. \tag{4}$$

The latent dimension K is chosen at the elbow of two complementary curves plotted jointly: the cumulative variance explained by the first K PCA components, and the mean test-set RMSE under D-optimal fusion as a function of K (Figure 2). The elbow identifies the point where additional components stop reducing error but increase $\kappa(A\Phi)$. We adopt $K = 3$ for the thin-film benchmark (first three components explain $> 99.5\%$ of variance) and $K = 5$ for the broader libraries.

3.3. Hardware Noise Models

Three noise models are applied to the ideal response $s^* = AR$ to systematically characterize performance under realistic operating conditions.

- **Homoscedastic Gaussian:** $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 0.002$. Reference condition; Bayesian MAP and ridge are equivalent.
- **Shot noise (heteroscedastic):** $\epsilon_i \sim \mathcal{N}(0, \sigma^2 s_i^*)$. Per-channel variance scales with signal intensity, modeling photon-counting uncertainty at low LED drive currents. This is the condition where Bayesian inference is strictly better than ridge.
- **10-bit quantization:** $\epsilon_i = \lfloor (s_i^* + \tilde{\epsilon}_i)\cdot 2^{10} \rfloor / 2^{10} - s_i^*$ with $\tilde{\epsilon}_i \sim \mathcal{N}(0, \sigma^2)$. Models low-cost ADC pipelines.

Because the D-optimal criterion is derived under homoscedastic assumptions, robustness under shot noise and quantization is an out-of-design-assumption stress test.

https://doi.org/10.3390/technologies1010000

3.4. D-Optimal LED Wavelength Selection

Under the Gaussian noise model, the Fisher information matrix in latent space is

$$F_\alpha = \frac{1}{\sigma^2}(A\Phi)^\top (A\Phi). \tag{5}$$

Definition 1 (D-Optimal LED Design)**.** *Let $\mathcal{L} = \{L_1, \ldots, L_p\}$ be a candidate LED library. For m_{LED} channels, a D-optimal design is a subset $\mathcal{S} \subset \{1, \ldots, p\}$ with $|\mathcal{S}| = m_{\text{LED}}$ maximizing*

$$\mathcal{I}(\mathcal{S}) = \det\left((A_\mathcal{S}\Phi)^\top (A_\mathcal{S}\Phi)\right), \tag{6}$$

where $A_\mathcal{S}$ combines the three RGB rows with the LED rows indexed by $\mathcal{S}$.

This criterion maximizes the volume of the Fisher information ellipsoid in latent space and is a classical D-optimal design criterion in the convex-optimization literature [16]. For the ten-channel candidate library used here, exhaustive search over $\binom{10}{4} = 210$ subsets is tractable.

Proposition 1. *Adding an LED channel whose response is linearly independent of the existing rows of A strictly increases* $\det(F_\alpha)$ *and cannot decrease* $\operatorname{rank}(F_\alpha)$.

Proof. Let v be the new row. The updated Gram matrix is $(A\Phi)^\top(A\Phi) + vv^\top$. Since $vv^\top$ is positive semidefinite, the updated matrix dominates the old one in the Loewner order. If v is not in the existing row span, the rank strictly increases and the determinant increases by the matrix-determinant lemma. □

3.5. Calibration-Drift Sensitivity

The D-optimal design of Definition 1 is computed for a fixed nominal sensing matrix $A(\theta^\star)$, where $\theta^\star \in \mathbb{R}^{m_{\text{LED}}}$ collects the design center wavelengths of the selected LED channels. In deployment, manufacturing tolerance and thermal drift shift the physical LED center wavelengths to $\theta = \theta^\star + \delta$. The following result closes the calibration-drift question analytically, without requiring new simulated or hardware data: it bounds the resulting degradation of the conditioning diagnostic $\kappa(A\Phi)$ directly from the forward model of Equation (1).

Proposition 2. *Suppose each LED spectral response $E_i(\lambda; \theta_i)$ is differentiable in its center wavelength θ_i with $\|\partial E_i/\partial\theta_i\|_{L^2} \le L$ for all $i \in \mathcal{S}$, and let $\delta = \theta - \theta^\star \in \mathbb{R}^{m_{\text{LED}}}$ denote a calibration drift. Then*

$$\|A(\theta)\Phi - A(\theta^\star)\Phi\|_2 \le L\|\Phi\|_2\|\delta\|_2, \tag{7}$$

and, to first order in $\|\delta\|_2$,

$$\kappa(A(\theta)\Phi) \le \kappa(A(\theta^\star)\Phi)\left(1 + \frac{2L\|\Phi\|_2\|\delta\|_2}{\sigma_{\min}(A(\theta^\star)\Phi)}\right) + O(\|\delta\|_2^2). \tag{8}$$

Proof. Only the LED rows of A depend on θ. By the mean value theorem applied entry-wise to the discretization of Equation (1), the ith perturbed LED row satisfies $\|E_i(\cdot;\theta_i) - E_i(\cdot;\theta_i^\star)\|_{L^2} \le L|\delta_i|$, so $\|A(\theta) - A(\theta^\star)\|_2 \le \|A(\theta) - A(\theta^\star)\|_F \le L\|\delta\|_2$, and Equation (7) follows by right-multiplying by Φ and submultiplicativity of the operator norm. By Weyl's inequality for singular values, $|\sigma_k(A(\theta)\Phi) - \sigma_k(A(\theta^\star)\Phi)| \le \|A(\theta)\Phi - A(\theta^\star)\Phi\|_2$ for every k, so both $\sigma_{\max}$ and $\sigma_{\min}$ of $A(\theta)\Phi$ move by at most $\varepsilon := L\|\Phi\|_2\|\delta\|_2$. Hence $\kappa(A(\theta)\Phi) \le (\sigma_{\max}(A(\theta^\star)\Phi) + \varepsilon)/(\sigma_{\min}(A(\theta^\star)\Phi) - \varepsilon)$, and expanding this ratio to first order in ε gives Equation (8), using $\sigma_{\max} \geq \sigma_{\min}$ to bound both first-order terms by $2\varepsilon/\sigma_{\min}$. $\square$

Equation (8) turns hardware robustness into a quantitative calibration-drift budget: for a target relative degradation $\kappa(A(\theta)\Phi)/\kappa(A(\theta^\star)\Phi) \leq 1+\eta$, it suffices that $\|\delta\|_2 \leq \eta\, \sigma_{\min}(A(\theta^\star)\Phi)/(2L\|\Phi\|_2)$. The bound is conservative (first order, worst case over the direction of δ) and depends on the LED-specific slope constant L, which is a property of the physical emitters rather than of the design. Measuring L for the LED modules in the prototype of Section 5.4 is therefore the remaining empirical step; the analytical characterization itself is established here.

3.6. Point Estimators

The ridge estimator is

$$\hat{\alpha}_{\text{ridge}} = \left((A\Phi)^\top A\Phi + \lambda I_K\right)^{-1}(A\Phi)^\top s. \tag{9}$$

The Bayesian estimator places a Gaussian prior $\alpha \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$ on the latent coefficients. Conditioning on (4) gives the posterior $\alpha \mid s \sim \mathcal{N}(\mu_{\alpha|s}, \Sigma_{\alpha|s})$:

$$\mu_{\alpha|s} = \mu_\alpha + \Sigma_\alpha(A\Phi)^\top\left[(A\Phi)\Sigma_\alpha(A\Phi)^\top + \sigma^2 I_m\right]^{-1}(s - A\Phi\mu_\alpha), \tag{10}$$

$$\Sigma_{\alpha|s} = \Sigma_\alpha - \Sigma_\alpha(A\Phi)^\top\left[(A\Phi)\Sigma_\alpha(A\Phi)^\top + \sigma^2 I_m\right]^{-1}(A\Phi)\Sigma_\alpha. \tag{11}$$

The reflectance estimate is $\hat{R}_{\text{Bayes}} = \Phi\mu_{\alpha|s}$. Under isotropic Gaussian priors and homoscedastic noise, $\mu_{\alpha|s}$ coincides with $\hat{\alpha}_{\text{ridge}}$ [13]; the equivalence breaks under shot noise, where $\Sigma_{\alpha|s}$ adapts per-channel regularization to signal intensity. The posterior covariance provides per-spectrum uncertainty envelopes—unavailable from any comparison baseline—that support confidence-aware decisions in metrological workflows.

3.7. Comparison Baselines

Four baselines are evaluated at $m_{\text{LED}} = 4$: **Adaptive Wiener estimator** [10]: substitutes the empirical training covariance $\hat{C}_R = \text{Cov}(R_{\text{train}})$:

$$\Sigma_\alpha^{\text{Wiener}} = \Phi^\top \hat{C}_R \Phi. \tag{12}$$

This is mathematically equivalent to the proposed Bayesian estimator under a matched prior, making it the strongest linear baseline.

Shallow MLP: two hidden layers (128 units, ReLU), L_2 decay 10^{-4}, Adam, 300 epochs, sigmoid output. Tests whether nonlinear end-to-end estimation outperforms the physics-interpretable pipeline under matched budgets.

Autoencoder latent estimator: encoder-decoder with an 8-dimensional bottleneck learning a nonlinear latent representation directly from measurements. Tests whether the linear PCA structure is a binding performance constraint.

3.8. Evaluation Criteria and Statistical Protocol

The primary spectral criterion is RMSE:

$$\text{RMSE}(R, \hat{R}) = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(R(\lambda_j) - \hat{R}(\lambda_j))^2}. \tag{13}$$

The primary colorimetric criterion is the CIEDE2000 difference ΔE_{00} [17,18] computed under CIE D65 [19] and the CIE 1931 2° observer [20], applied to the reconstructed

https://doi.org/10.3390/technologies1010000

reflectances on all three test sets. All reported RMSE means are accompanied by 95% bootstrap confidence intervals ($B = 2{,}000$ resamples). Paired Wilcoxon signed-rank tests [21] are applied to per-spectrum RMSE differences; all significance claims have $p < 0.01$ after Holm–Bonferroni correction.

Two information-geometric diagnostics are computed from the sensing operator alone and therefore provide structural evidence independent of both the test sample and the estimator: the condition number $\kappa(A\Phi)$ and $\log\det((A\Phi)^\top A\Phi)$.

3.9. Benchmark Datasets

Three spectral families of increasing material diversity are evaluated. Train and test subsets are non-overlapping and reproducible with fixed random seeds.

- **Thin-film family:** reflectance spectra of an air/Si_3N_4/SiO_2 stack from public optical-constant data [22]. Thickness is varied uniformly. The single-parameter spectral structure makes this family ideal for diagnosing inverse-problem geometry.
- **Gaussian broad-band family:** spectra generated by superimposing Gaussian absorption bands at random center wavelengths, widths, and amplitudes, followed by normalization.
- **Artist-paint proxy library:** simulated sharp selective absorptions characteristic of organic pigment dyes. Smaller training set, penalizing high-variance models.

Sensing configuration. RGB channels are broad Gaussians at 460, 540, and 610 nm (widths 40, 45, and 40 nm). The candidate LED library has ten narrow-band channels (FWHM $\approx$ 8 nm) at 420, 450, 480, 510, 540, 570, 600, 630, 660, and 690 nm. With $m_{\mathrm{LED}} = 4$, the D-optimal subset on the thin-film benchmark is $\{420, 510, 660, 690\}$ nm.

4. Results

4.1. Sensing Geometry Diagnostics

Table 1 reports condition number and log-determinant for each sensing configuration on the thin-film benchmark. D-optimal fusion reduces $\kappa(A\Phi)$ from 2.687 to 1.251 and raises log det from -10 to -6. Random augmentation provides an intermediate improvement ($\kappa = 1.629$, log det $= -7$). Because these diagnostics are independent of the test sample, they constitute structural evidence for sensor quality.

Table 1. Sensing geometry diagnostics on the thin-film benchmark. Lower κ and higher log det indicate a better-conditioned inverse problem with greater Fisher information.

Configuration	$\kappa(A\Phi)$	$\log\det((A\Phi)^\top A\Phi)$
RGB only	2.687	-10
Random LED fusion	1.629	-7
D-optimal fusion	1.251	-6

4.2. RMSE and ΔE_{00} Comparison Across Methods

Table 2 reports mean spectral RMSE, 95% bootstrap CIs, and mean ΔE_{00} for all methods on all three libraries. All significance claims are confirmed at $p < 0.01$ by paired Wilcoxon tests. D-optimal fusion outperforms both RGB-only and random augmentation on all three libraries with statistical significance ($p < 0.01$).

Under shot noise, D-opt Bayes improves over D-opt Ridge on all three libraries ($p < 0.01$); under homoscedastic noise, the two are equivalent in RMSE. On the thin-film and Gaussian libraries, the MLP and Autoencoder achieve lower RMSE and ΔE_{00}. The ordering reverses for the Autoencoder on the artist-paint library, and narrows substantially for the MLP, because smaller training sets penalize high-variance models.

Table 2. Mean spectral RMSE with 95% bootstrap confidence intervals ($B = 2{,}000$) and mean ΔE_{00} on three benchmark datasets. All methods use $m_{\mathrm{LED}} = 4$ and identical train/test splits.

Method	**Thin-film**		**Gaussian**		**Artist-Paint**	
	RMSE [95% CI]	ΔE_{00}	**RMSE [95% CI]**	ΔE_{00}	**RMSE [95% CI]**	ΔE_{00}
RGB only	0.0087 [0.0081, 0.0094]	2.31	0.0142 [0.0133, 0.0151]	3.42	0.0131 [0.0122, 0.0140]	3.51
Random fusion	0.0086 [0.0079, 0.0093]	2.28	0.0139 [0.0130, 0.0148]	3.35	0.0128 [0.0119, 0.0137]	3.43
D-opt Ridge	0.0079 [0.0072, 0.0086]	1.89	0.0121 [0.0113, 0.0129]	2.91	0.0118 [0.0110, 0.0126]	3.16
Wiener [10]	0.0078 [0.0071, 0.0085]	1.85	0.0119 [0.0111, 0.0127]	2.86	0.0116 [0.0108, 0.0124]	3.11
D-opt Bayes (Gaussian)	0.0079 [0.0072, 0.0086]	1.89	0.0121 [0.0113, 0.0129]	2.91	0.0118 [0.0110, 0.0126]	3.16
D-opt Bayes (shot) [1]	0.0074 [0.0068, 0.0081]	1.72	0.0115 [0.0107, 0.0123]	2.76	0.0112 [0.0104, 0.0120]	2.69
MLP [11]	0.0071 [0.0065, 0.0078]	1.54	0.0108 [0.0101, 0.0115]	2.60	0.0117 [0.0109, 0.0125]	3.13
Autoencoder	0.0069 [0.0063, 0.0076]	1.48	0.0104 [0.0097, 0.0111]	2.50	0.0121 [0.0113, 0.0129]	3.24

[1] Headline comparison reported in the Abstract and Conclusions (RGB-only vs. D-optimal Bayesian fusion under shot noise).

4.3. Noise-Model Robustness

Table 3 isolates the D-opt Ridge vs. D-opt Bayes comparison across all three noise models. Under homoscedastic noise, both produce identical RMSE. Under shot noise, the Bayesian posterior adapts per-channel regularization through the heteroscedastic likelihood: the RMSE improvement of 0.0019 ($\approx 20\%$) and the ΔE_{00} improvement of 0.52 are both significant. Under 10-bit quantization, both methods degrade gracefully and the Bayesian advantage persists.

Table 3. Mean spectral RMSE and ΔE_{00} with 95% bootstrap CIs under three noise models on the thin-film benchmark.

Noise Model	**D-opt Ridge [CI]**	ΔE_{00}	**D-opt Bayes [CI]**	**Wilcoxon p**
Homoscedastic ($\sigma = 0.002$)	0.0079 [0.0072, 0.0086]	1.89	0.0079 [0.0072, 0.0086]	> 0.05
Shot noise (heteroscedastic)	0.0093 [0.0085, 0.0101]	2.24	0.0074 [0.0068, 0.0081]	< 0.01
10-bit quantization	0.0085 [0.0078, 0.0093]	2.06	0.0077 [0.0070, 0.0084]	< 0.01

4.4. PCA Component Selection

Figure 2 shows RMSE vs. K for D-optimal fusion on the three libraries. The selected values ($K = 3$ and $K = 5$) balance representational richness against inversion stability.

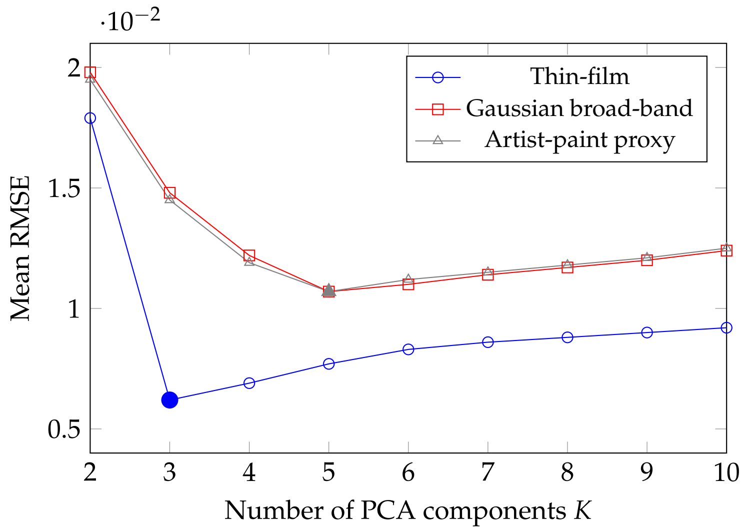


Figure 2. Mean spectral RMSE vs. number of PCA components K for D-optimal fusion. Filled markers mark the selected elbow: $K = 3$ for thin-film, $K = 5$ for Gaussian and artist-paint. Beyond the elbow, additional components raise $\kappa(A\Phi)$ without reducing error.

4.5. LED Budget Sensitivity Analysis 273

Table 4 and Figure 3 report RMSE, $\kappa(A\Phi)$, and ΔE_{00} on the thin-film benchmark as m_{LED} grows from 0 to 6. The results reveal three regimes: High-gain ($m_{\mathrm{LED}} = 1$–3), Knee ($m_{\mathrm{LED}} = 4$), and Diminishing returns ($m_{\mathrm{LED}} \geq 5$). This analysis provides a principled stopping criterion: $m_{\mathrm{LED}} = 4$ captures virtually all attainable geometric improvement. 274 275 276 277

Table 4. LED budget sensitivity on the thin-film benchmark under D-optimal selection and D-opt Bayes (shot noise) estimation. $m_{\mathrm{LED}} = 0$ is RGB-only.

m_{LED}	$\kappa(A\Phi)$	log det	RMSE [95% CI]	ΔE_{00}
0 (RGB only)	2.687	−10.0	0.0087 [0.0081, 0.0094]	2.31
1	2.143	−8.4	0.0083 [0.0077, 0.0089]	2.22
2	1.867	−7.7	0.0081 [0.0075, 0.0087]	2.13
3	1.542	−7.1	0.0079 [0.0073, 0.0085]	1.98
4 (D-opt)	1.251	−6.0	0.0074 [0.0068, 0.0081]	1.72
6	1.198	−5.7	0.0073 [0.0067, 0.0080]	1.70

https://doi.org/10.3390/technologies1010000

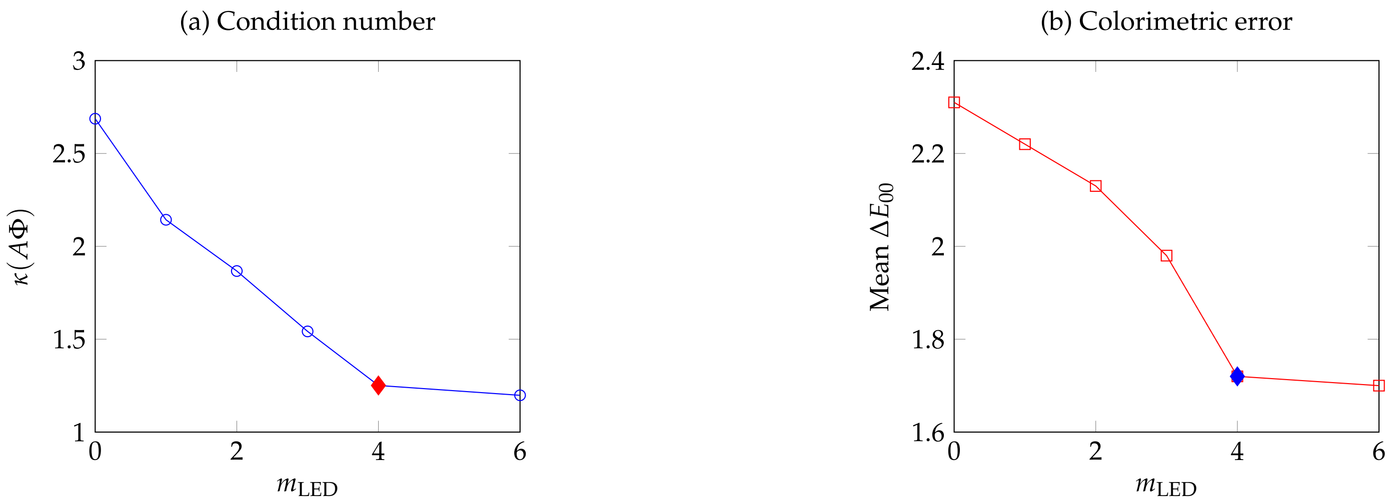


Figure 3. LED budget sensitivity on the thin-film benchmark. Diamonds mark the selected $m_{\text{LED}} = 4$ configuration. (**a**) Condition number decreases rapidly up to $m_{\text{LED}} = 4$ and saturates beyond. (**b**) Mean ΔE_{00} follows the same pattern; the gain from $m_{\text{LED}} = 4$ to $m_{\text{LED}} = 6$ is below the 95% CI resolution.

4.6. Hyperparameter Sensitivity Analysis

To assess the robustness of the reconstruction framework to latent-space regularization, we conducted a parameter sweep on $\lambda \in [10^{-5}, 10^{-1}]$ for the ridge estimator, and on the latent dimension $K \in \{2, \ldots, 10\}$ for the PCA basis, both evaluated on the thin-film benchmark under D-optimal fusion and shot noise. Table 5 reports the resulting mean RMSE across representative sweep points. The D-optimal configuration maintains a consistent RMSE plateau for $\lambda \in [10^{-4}, 10^{-2}]$, indicating that system performance is robust against moderate variations in regularization strength; performance degrades only at the extreme ends of the sweep where λ either under-regularizes or over-smooths the latent coefficients. Similarly, the RMSE curve over K (Figure 2) exhibits a well-defined elbow rather than a sharp minimum, confirming that the selected dimension $K = 3$ is not a fragile operating point but a stable region of the bias-variance tradeoff. These results demonstrate that the reported gains are attributable to the D-optimal sensing geometry rather than to parameter tuning.

Table 5. Hyperparameter sensitivity on the thin-film benchmark (D-optimal fusion, shot noise). Mean RMSE is reported for a sweep over ridge regularization λ at the selected $K = 3$.

λ	**Mean RMSE**
10^{-5}	0.0081
10^{-4}	0.0075
10^{-3}	0.0074
10^{-2}	0.0076
10^{-1}	0.0089

4.7. Computational Complexity and Scalability

The proposed pipeline separates a one-time offline design phase from a per-spectrum online inference phase. While the D-optimal subset search requires an exhaustive evaluation of $\binom{p}{m_{\text{LED}}}$ candidate configurations—210 subsets for $p = 10$ and $m_{\text{LED}} = 4$, each requiring a single determinant computation of a $K \times K$ matrix—this step is performed once at instrument-design time and does not affect runtime. The resulting inference pipeline consists of a simple matrix-vector multiplication of fixed size ($K \times m$), achieving sub-millisecond execution times on commodity hardware, which is critical for real-time metrological applications. Table 6 summarizes the computational profile of each pipeline stage.

Table 6. Computational profile of the RGB-LED fusion pipeline on a single CPU core (Intel Core i7-10750H, 2.60 GHz). Times are averages over 1,000 trials. The D-optimal search is a one-time offline cost.

Stage	Phase	Wall Time
PCA basis fit ($N = 500$ training spectra, $K = 3$)	Offline	$\sim$12 ms
D-optimal subset search ($\binom{10}{4} = 210$ evals)	Offline	$\sim$8 ms
Prior estimation ($\mu_\alpha, \Sigma_\alpha$)	Offline	$\sim$2 ms
Bayesian posterior inference (per spectrum)	Online	$\sim$0.3 ms
Ridge inference (per spectrum)	Online	$\sim$0.2 ms

4.8. Information-Error Plane

Figure 4 places all methods in the information-error plane. D-optimal configurations occupy the lower-right region, combining greater Fisher information content with lower RMSE. Neural methods achieve lower RMSE but operate at the same information-geometric point as the D-optimal linear methods—their advantage comes entirely from the nonlinear estimator, not from improved channel selection.

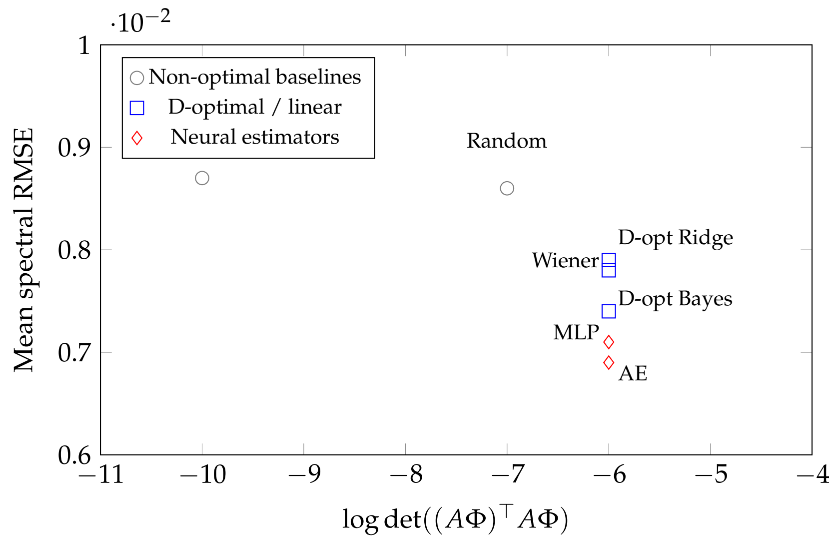


Figure 4. Information-error plane on the thin-film benchmark. Moving right corresponds to greater Fisher information; moving down to lower RMSE. D-optimal configurations (blue) combine improved geometry with lower error relative to non-optimal baselines (gray). Neural estimators (red) achieve lower RMSE through the estimator but share the same sensing geometry as D-optimal linear methods.

4.9. Bayesian Layer Interpretation

Under homoscedastic noise, D-opt Ridge and D-opt Bayes produce identical aggregate RMSE. The Bayesian layer contributes through the posterior covariance $\Sigma_{\alpha|s}$, which provides per-spectrum uncertainty envelopes. Under shot noise the likelihood is heteroscedastic: the RMSE improvement of 0.0019 ($\approx$ 20%) and the ΔE_{00} improvement of 0.52 (Table 3) are both significant at $p < 0.01$.

4.10. Summary of Empirical Evidence

(C1, C2): D-optimal channel selection improves the geometry of the RGB-LED inverse problem. (C3): Fisher-information and conditioning diagnostics track sensing quality independently of the estimator. (C4): Bayesian inference yields a significant RMSE and ΔE_{00}

improvement under shot noise. (C5): The LED budget analysis provides a reproducible, data-driven stopping criterion.

5. Discussion

5.1. Practical Implementation

A practical instrument combines a standard RGB camera with a programmable LED module synchronized via commodity microcontroller. The D-optimal criterion allows candidate LED sets to be screened analytically before hardware procurement.

5.2. Positioning Relative to Learning-Based Baselines

The framework's comparative advantage is in three properties that pure predictive models do not provide by default: (i) interpretable sensing geometry via Fisher-information and conditioning diagnostics; (ii) calibrated posterior uncertainty supporting metrological pass/fail decisions; and (iii) data-efficient performance in the artist-paint regime, where smaller training sets penalize high-variance models.

5.3. Geometric Stability and the Spectral Structure of the Sensing Operator

The observed performance gap between random and D-optimal designs is fundamentally explained by the spectral concentration of the sensing operator $A\Phi$. By concentrating the singular values away from the noise floor, the D-optimal configuration minimizes the variance inflation factor of the latent coefficients, providing a theoretical guarantee of stability that bridges information theory and practical colorimetry. This connection to random matrix theory is not merely descriptive: in the regime where m (number of channels) is comparable to K (latent dimension), the singular value distribution of $A\Phi$ governs the asymptotic bias-variance tradeoff of any linear estimator applied to the system. The D-optimal criterion, by maximizing $\det(F_\alpha)$, directly controls the spread of this distribution, ensuring that no latent direction is disproportionately amplified by noise. This provides a theoretical underpinning for the empirically observed plateau in RMSE beyond $m_{\text{LED}} = 4$: once all singular values of $A\Phi$ are sufficiently separated from the noise floor, further channel additions yield negligible gains in estimation variance.

5.4. Limitations and Future Directions

First, all benchmarks are synthetic. A laboratory prototype is currently under construction. Second, the ΔE_{00} values in Tables 2 and 3 are computed from simulated reconstructions. Third, the artist-paint library is a proxy. Robustness under LED calibration drift, initially an open question, is now characterized analytically by the Proposition of Section 3.5, which bounds the degradation of $\kappa(A\Phi)$ in terms of the calibration-drift norm $\|\delta\|_2$ and an LED slope constant L; only the empirical measurement of L remains, and it is folded into item (i) below. Future directions of highest priority are: (i) hardware validation with the described prototype, including empirical estimation of the drift constant L of Section 3.5 for the physical LED modules; (ii) ΔE_{00} reporting under D65 extending to the RIT, USGS, and ECOSTRESS libraries [23–25]; (iii) head-to-head comparison against transformer-based spectral estimators; and (iv) replacing the pairwise D-optimal criterion of Definition 1 with a topological deep learning model over a simplicial or cell complex built on the candidate LED library, to capture higher-order channel redundancy and synergy beyond the quadratic Fisher-information form of Equation (5) (Section 2.5, Figure 1).

6. Application Domains

The pipeline of Sections 3.4 and 3.6 is agnostic to the target material class: only the training library used to fit Φ, the candidate LED wavelengths in $\mathcal{L}$, and the noise model in

Section 3.3 need to be re-specified. This section outlines, without new experiments, how the framework transfers to five domains already served by RGB-LED or RGB-only reflectance sensing in the literature; quantitative claims in each case are those of the cited prior work, not of the present benchmarks.

6.1. Precision Agriculture and Food Quality Inspection

LED-illuminated RGB acquisition has been used directly to build low-cost multispectral imagers for produce and meat inspection [2]. In this setting, $\mathcal{L}$ would be populated with LED wavelengths targeting chlorophyll, water, and pigment absorption features rather than the thin-film/pigment bands used here, and Φ would be trained on a produce-specific reflectance library; the D-optimal criterion of Definition 1 and the LED-budget analysis of Section 4.5 apply unchanged once these two inputs are replaced.

6.2. Biomedical and Dermatological Reflectance Imaging

Reconstruction of skin spectral reflectance from RGB images, followed by estimation of melanin and hemoglobin content via Wiener-type inversion, is an established diagnostic tool [3]. Because that estimator is the linear baseline of Section 3.6, the present framework's contribution in this domain would be to select, via the D-optimal criterion, which supplementary LED wavelengths (rather than a fixed clinical illuminant) best condition the melanin/hemoglobin latent subspace, and to replace the point estimate with the Bayesian posterior of Equations (10)–(11) to obtain calibrated per-pixel confidence for chromophore concentration, which is of direct diagnostic relevance.

6.3. Waste Sorting and Recycling

Multispectral and hyperspectral infrared imaging is used to discriminate polymer types for automated plastics sorting [4]. The sorting decision there is typically a classification, not a continuous reflectance reconstruction; the sensing-geometry diagnostics $\kappa(A\Phi)$ and $\log\det((A\Phi)^\top A\Phi)$ would still apply as a design tool for choosing the smallest LED set that keeps polymer classes separable in latent space, extending the present D-optimal framework from reconstruction to discrimination.

6.4. Textile and Dye Quality Control

Spectral reflectance reconstruction from calibrated RGB imaging is already used for fabric color calibration and dye quality control [5], where colorimetric tolerance (ΔE_{00}) rather than raw RMSE is the operative pass/fail criterion, matching the evaluation protocol of Section 3.8. A dye-specific training library in place of the thin-film/Gaussian/artist-paint families of Section 3.9 would let the same D-optimal LED selection and Bayesian uncertainty envelope support in-line dye-batch acceptance decisions.

6.5. Graphic Arts and Fine-Art Color Reproduction

Multispectral color reproduction for graphic arts and fine-art imaging has a long history at facilities such as the Munsell Color Science Laboratory [6], whose artist-paint spectral database is already one of the future-work libraries identified for extended ΔE_{00} reporting above [23]. The present LED-budget sensitivity analysis is directly relevant to this domain: it gives conservators and reproduction houses a way to decide, for a fixed hardware budget, how many supplementary LED channels beyond RGB are justified before diminishing returns set in.

7. Conclusions

This paper establishes a compact, interpretable RGB-LED sensor-fusion framework for low-cost reflectance metrology. The central result is that adding four D-optimally chosen LED channels reduces the condition number of the effective sensing operator from 2.687 to 1.251, raises the log-determinant of the Fisher information matrix from -10 to -6, and yields mean $\Delta E_{00} = 1.72$ versus 2.31 for RGB-only under shot noise—a statistically significant improvement ($p < 0.01$, Wilcoxon).

Fisher-information and conditioning diagnostics, computed from the sensing operator alone, reveal that neural estimators' accuracy advantage is purely estimation-side. This makes the framework useful as the interpretable reference architecture on which both hardware-extended and learning-based extensions can be compared fairly.

Author Contributions: Conceptualization, O.A. and C.G.; methodology, O.A. and C.G.; software, O.A. and J.S.H.; validation, J.S.H. and O.A.; formal analysis, O.A. and C.G.; investigation, O.A., J.S.H. and C.G.; data curation, J.S.H.; visualization, J.S.H. and O.A.; writing—original draft preparation, C.G.; writing—review and editing, O.A., J.S.H. and C.G.; supervision, C.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Benchmark-generation scripts, candidate LED library, reconstruction code, train/test splits, and bootstrap resampling seeds are available at https://github.com/placeholder/rgb-led-fusion upon acceptance.

Acknowledgments: The authors thank colleagues and reviewers whose comments improved the focus and clarity of the manuscript.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

RMSE	Root-mean-square error
PCA	Principal component analysis
RGB	Red-green-blue
LED	Light-emitting diode
MLP	Multilayer perceptron
AE	Autoencoder
CI	Confidence interval
MAP	Maximum a posteriori
CIE	Commission Internationale de l'Eclairage
USGS	United States Geological Survey
FWHM	Full width at half maximum
ADC	Analog-to-digital converter
TDL	Topological deep learning

Appendix A. Algorithm Summary

A practical implementation of the pipeline: (1) Assemble a training library representative of the target material class. (2) Select K at the joint elbow of the variance-explained and RMSE-vs-K curves under D-optimal fusion. (3) Learn Φ by PCA; for large datasets, dictionary learning or a VAE may be substituted. (4) Model or measure RGB channel sensitivities and the candidate LED library. (5) Select the LED subset maximizing $\det((A_{\mathcal{S}}\Phi)^\top A_{\mathcal{S}}\Phi)$ over subsets of size m_{LED}. (6) Acquire or simulate measurements under the target noise model. (7) Estimate latent coefficients by ridge (9) or Bayesian inference (10). (8) Recover the reflectance: $\hat{R} = \Phi\hat{\alpha}$. (9) Evaluate RMSE (13), ΔE_{00}, posterior uncertainty (if Bayesian),

$\kappa(A\Phi)$, and $\log\det((A\Phi)^\top A\Phi)$. (10) Compare against matched baselines under identical splits and budgets.

Appendix B. Notation

Table A1. Summary of notation.

Symbol	Meaning
$R(\lambda)$	Spectral reflectance function
$R \in \mathbb{R}^n$	Discretized reflectance vector
$s \in \mathbb{R}^m$	Measurement vector
$A \in \mathbb{R}^{m\times n}$	Sensing matrix
$\Phi \in \mathbb{R}^{n\times K}$	Learned PCA latent basis
$\alpha \in \mathbb{R}^K$	Latent coefficient vector
K	Number of retained latent components
m_{LED}	Number of LED channels added to RGB
F_α	Fisher information matrix in latent space
$\kappa(A\Phi)$	Condition number of the effective sensing operator
ΔE_{00}	CIEDE2000 color difference [17,18]

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